

Formalizing Copyright Law Through Melody Similarity

An explainable decision-support prototype for music copyright disputes

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Project claim. We do not automate copyright adjudication. We make one legally relevant issue explicit and inspectable: whether two songs share a recognizable melodic fragment.

1 Legal Problem

Music copyright disputes often turn on the open-textured standard of **substantial similarity**. But similarity is not itself a legal conclusion.

- Courts also consider access, protectability, scope of the disputed material, and context.
- Raw similarity scores can be persuasive but legally misleading if their meaning is hidden.
- The research gap is not only computational; it is techno-legal: how can a score support legal reasoning without pretending to replace it?

Research question. Can local melodic alignment capture legally relevant fragment similarity better than global comparison while remaining explainable enough for decision support?

2 Achieved Scope

The fall pitch proposed a broad formalization pipeline. The implemented poster-stage prototype focuses on the part that can be defended with the current code and data.

Initial ambition	Achieved component
Legal formalization	Minimal legal variables and rule-trace logic.
Similarity layer	Needleman–Wunsch global alignment, Smith–Waterman local alignment, and PMI scores.
Threshold analysis	Exploratory thresholds on 15 melody-dispute pairs.
Decision support	Scorecard interpretation, not automated judgment.

Why this is credible. The poster presents the working evidence layer clearly: what it measures, where it helps, and where legal judgment must remain in control.

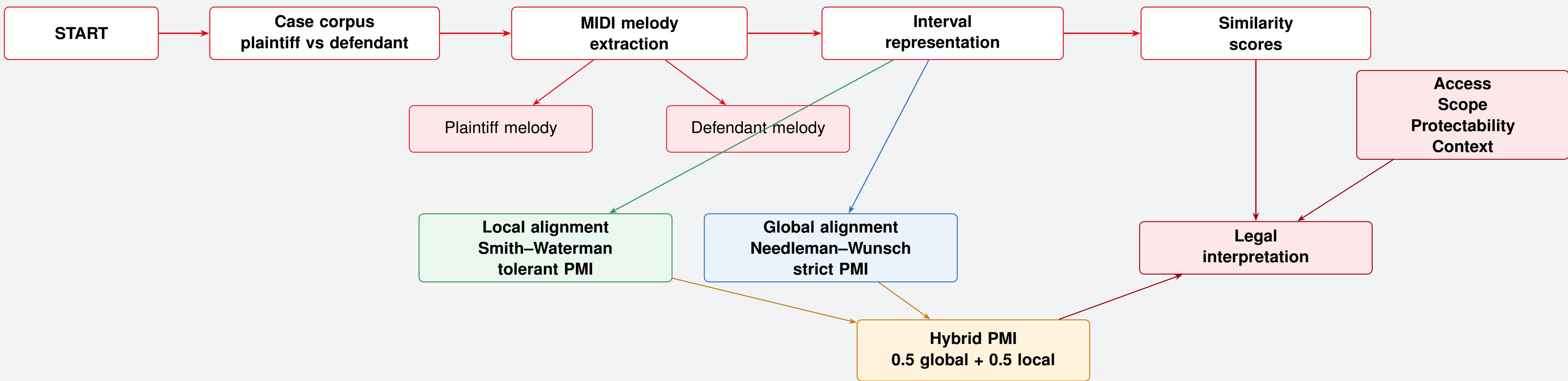
3 Dataset and Legal Coding

Artifact	Current state
Corpus	128 music copyright disputes in the master case file.
Evaluation subset	15 pairs with curated MIDI melody excerpts.
Representation	Monophonic pitch sequences converted to intervals.
Legal variables	Access, melody scope, protectability, context override.

Bounded contribution. The project is strongest when presented as an explainable similarity and legal-interpretation layer for melody-focused cases.

4 Method Overview

Implemented pipeline. MIDI melody excerpts are transformed into interval sequences, aligned globally and locally, then interpreted through a minimal legal rule layer.



Melody representation.

$$I = [p_2 - p_1, p_3 - p_2, \dots, p_n - p_{n-1}]$$

Intervals make comparison key-invariant and focused on melodic contour.

Alignment layer.	
Global	Aligns the whole interval sequences and penalizes gaps: useful for overall resemblance.
Local	Finds the best matching subsequence: useful when alleged copying is a recognizable fragment.

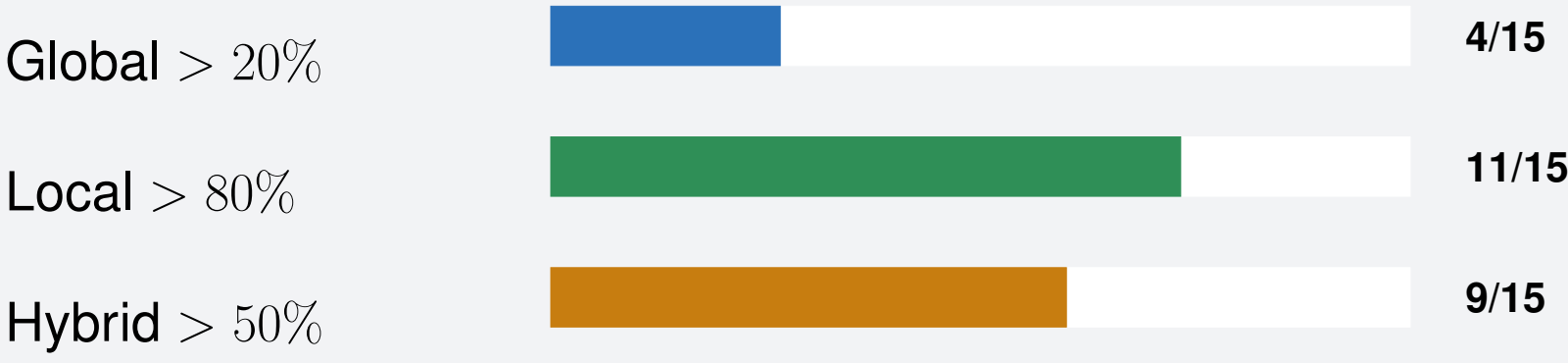
Global PMI
Whole-excerpt resemblance. Conservative, but rigid when copying is fragment-based.

Local PMI
Best-matching fragment resemblance. Closer to legally relevant motif copying.

Hybrid PMI
Balances whole-melody and fragment evidence so one local match does not decide the case alone.

5 Results

Threshold behavior on 15 curated melody pairs.



Main empirical finding. Local PMI detects substantially more known fragment-based disputes than global PMI.

Legal caution. High similarity is over-inclusive: a score needs access, protectability, scope, and context before it becomes legally meaningful.

Illustrative readings.		
Pair	G	L
No Scrubs / Shape of You	8.39	92.12
Bitter Sweet Symphony / The Last Time	26.95	77.54
Ice Ice Baby / Under Pressure	2.78	33.33
Flowers / When I Was Your Man	36.10	89.43

Full score table. Global and local scores are shown separately; the hybrid score is their simple average.

#	Melody pair	Global	Local	Hybrid
1	The Last Time / Bitter Sweet Symphony	26.95	77.54	52.24
2	Anybody Seen My Baby / Constant Craving	5.01	93.08	49.04
3	All You Need Is Love / In the Mood	12.85	95.05	53.95
4	Oops Up Side Your Head / Uptown Funk	6.58	62.90	34.74
5	Greatest Love of All / If You Could Read My Mind	21.10	93.23	57.16
6	Sweet Little Sixteen / Surfin' U.S.A.	18.62	93.70	56.16
7	I Won't Back Down / Stay With Me	7.76	88.89	48.32
8	No Scrubs / Shape of You	8.39	92.12	50.25
9	I'm Not Alone / Yeah 3x	3.61	0.00	1.81
10	Teach the World to Sing / Shakermaker	12.90	89.46	51.18
11	Under Pressure / Ice Ice Baby	2.78	33.33	18.06
12	Every Breath You Take / I'll Be Missing You	23.01	93.35	58.18
13	All Day and All of the Night / Hello, I Love You	12.74	93.94	53.34
14	I Want a New Drug / Ghostbusters	9.91	85.00	47.45
15	When I Was Your Man / Flowers	36.10	89.43	62.77

Interpretation. Global PMI is useful as a conservative baseline; local PMI is better aligned with the legal intuition that a recognizable fragment can matter even when the full melodies differ.

Global result. Conservative whole-excerpt comparison flags only 4/15 pairs, so it misses several fragment-centered disputes.

Local result. Best-fragment comparison flags 11/15 pairs and better matches the way melody disputes are argued.

Hybrid result. The combined score flags 9/15 pairs, keeping sensitivity while avoiding reliance on one fragment alone.

6 Legal Interpretation

The contribution is not that scores decide cases. The contribution is a transparent structure for interpreting scores.

Factor	Role
Access	Similarity is weaker without plausible access.
Melody scope	The tool should not decide lyrics, timbre, production, or vibe cases.
Protectability	High similarity can still concern generic material.
Context	Unusual cases require manual legal review.

Rule-trace logic. Stop if outside melodic scope; lower concern if access is absent; flag review only when similarity and legal factors align.

7 Discussion

- Empirical:** local alignment better captures recognizable fragments than full-excerpt comparison.
- Legal:** similarity is evidence, not the legal test itself.
- Computational:** the model is interpretable because each score has a clear methodological meaning.
- Capstone value:** the work connects a measurable artifact to a legal reasoning structure.

Takeaway. Computation can clarify one part of copyright reasoning, but it should support human legal judgment rather than replace it.

8 Conclusion

Final claim. The project achieved a defensible part of the original capstone goal: an explainable computational layer for melody-based copyright analysis.

Limitation	Next step
Melody only	Add rhythm, harmony, sampling, production.
15 MIDI pairs	Expand and balance evaluation.
Settlements as labels	Separate settlements from court findings.
Exploratory thresholds	Calibrate and report uncertainty.
Legal variables incomplete	Finish annotations and rule traces.

Final takeaway. The prototype turns melodic similarity from a black-box number into auditable evidence: useful for legal reasoning, but never a substitute for legal judgment.